

Petar Nizić-Nikolac

Born	2000, Zagreb, Croatia
Location	HG G 21.1, Rämistrasse 101, Zürich, Switzerland
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CV update	10/2024

Education

PhD in Mathematics	ETH Zürich • Supervisor: Prof. Afonso Bandeira , Second advisor: Prof. Vincent Tassion • Interests: Random Matrices, High Dimensional Probability, Statistical Learning, Combinatorics, Optimization, Percolation	10/2022 - <i>ongoing</i>
MMath (Master of Mathematics)	University of Cambridge, Trinity College • Awarded Honours with Distinction	09/2021 - 06/2022
BA (Bachelor of Arts) in Mathematics	University of Cambridge, Trinity College • Awarded First Class in each examination session	09/2018 - 06/2021
Mathematical grammar school	XVth Gymnasium Zagreb • Awarded prize for the best student of generation 2018	09/2014 - 06/2018

Work Experience

Doctoral Student	ETH Zürich • Teaching Assistant – Fall/2024: Mathematics of Data Science (course organizer) – Spring/2024: Mathematics of Signals, Networks, and Learning (course organizer) – Fall/2023: Mathematics of Data Science – Spring/2023: Mathematics of Signals, Networks, and Learning • Student Supervision – Spring/2024: Mael Macuglia (MSc thesis), Stefan Hadzhistoykov (BSc thesis) – Fall/2023: Nicolas Hotton (semester project) – Spring/2023: Xinmeng Zeng (MSc thesis), Danielle Ludin (semester project) • Organization – DACO Seminar, 10/2023 - <i>ongoing</i> • Talks and Posters – Mathematical Aspects of Learning Theory - 20 years later, Barcelona, 09/2024 – Research Seminar Noncommutative and Functional Analysis, Saarland, 11/2023 – Graduate Seminar in Probability, ETH Zürich, 12/2022	10/2022 - <i>ongoing</i>
Quant Software Engineer Intern	Squarepoint Capital (London) • Working in High Performance Quantitative Software Engineering team during summer • Designed a system to detect and present unusual behaviours of the market	06/2021 - 08/2021
Summer Research Intern	University of Cambridge, Faculty of Mathematics • Algorithmic-deterministic lower bounds on the percolation threshold for the grid $p_c(\mathbb{Z}^d)$ • Researching the bunkbed conjecture and proving a limiting result • T. Hutchcroft, A. Kent, P. Nizić-Nikolac, The bunkbed conjecture holds in the $p \uparrow 1$ limit , Combinatorics, Probability and Computing 2022, 1–7.	06/2020 - 09/2020

Outreach

Member	Croatian Mathematical Society • Team Selection Test Committee • Deputy Leader at 16 th Middle European Mathematical Olympiad, Switzerland, 2022	09/2021 - <i>ongoing</i>
Member	Society Young Gifted Mathematicians 'Marin Getaldić' • European Mathematical Cup Jury, organizing camps and popularizing mathematics	07/2018 - 08/2022
Junior Treasurer	Cambridge University Croatian Society	09/2020 - 06/2022
Instructor	XVth Gymnasium Zagreb • Preparing students from former high school for national competitions	09/2018 - 04/2019

Awards and Competitions

University Awards	Scholarships • Senior Scholar Award, Cambridge, <i>2021</i> • Junior Scholar Award, Cambridge, <i>2019</i> Studentships • F. J. Woods Fund Studentship, Cambridge, <i>2020</i> • CMS Summer Studentship, Cambridge, <i>2020</i>
Competitions	Mathematical Olympiads • Bronze medal at <i>59th</i> International Mathematical Olympiad, Romania, <i>2018</i> • Bronze medal at <i>58th</i> International Mathematical Olympiad, Brasil, <i>2017</i> • 2×Gold medal at <i>10th</i> Middle European Mathematical Olympiad, Austria, <i>2016</i> • Gold+Silver medal at <i>9th</i> Middle European Mathematical Olympiad, Slovenia, <i>2015</i> Other International Competitions • Silver medal at <i>12th</i> International Junior Science Olympiad, South Korea, <i>2015</i> • <i>1st</i>, <i>2nd</i>, <i>2nd</i> place on American Computer Science League, USA, <i>2015</i>, <i>2016</i>, <i>2017</i>

Knowledge and Skills

Languages	English (fluent), Croatian (native), German (basic)
Computer skills	C++ , Python , MATLAB , Q/kdb+ , Latex , GitHub